MTH 202 Chapter 5: Topics In Vector Calculus

Applications of Surface Integrals; Flux Worksheet

1. Let σ be the portion of the cylinder $x^2 + y^2 = 16$ between the planes $z = -2$ and $z = 2$ oriented by outward normal. Compute the flux of **F** across σ, where $\mathbf{F}(x, y, z) = \langle 3, -7, z \rangle$.

2. Let σ be the portion of the plane $6x+3y+2z=6$ in the first octant oriented by upwards normal and $\mathbf{F}(x,y,z)=\langle x^2, yx, zx\rangle$. Compute the flux of **F** across σ in the positive direction.

3. Let σ be the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 4$ oriented by inward normals and $\mathbf{F}(x, y, z) = \mathbf{i} + y\mathbf{j} + \mathbf{k}$. Compute the flux of $\mathbf{F}$ across σ in the direction of orientation.

4. Let σ be the portion of the sphere $x^2 + y^2 + z^2 = 1$ that is inside the cone $z = \sqrt{\frac{1}{3}x^2 + \frac{1}{3}y^2}$ and $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Find the flux of **F** across σ in the positive direction.

5. Let $\mathbf{F}(x, y, z) = 2xy\mathbf{i} - y^2\mathbf{j} + 2z^3\mathbf{k}$. Find the inward flux of the surface that is bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 2$.